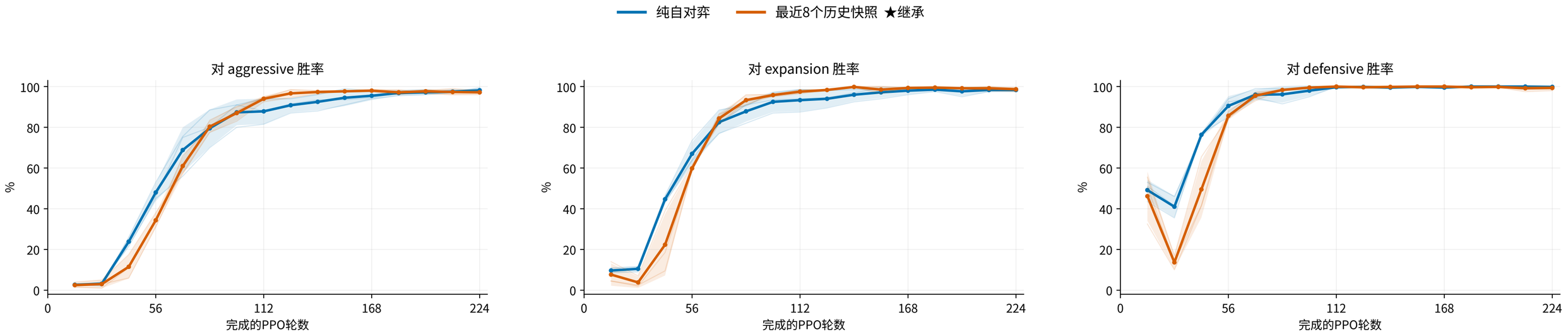
S1 对手机制：模型能力随训练变化

3个配对种子｜224轮PPO｜每轮256次逻辑Adam/rank｜周期评估每14轮一次、每次800局

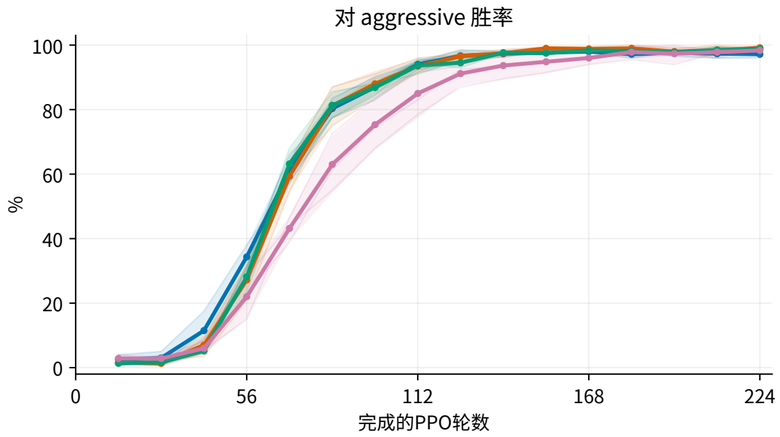


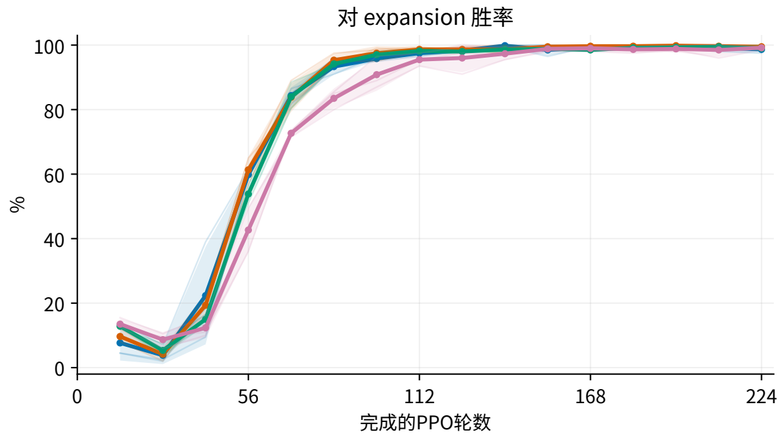
粗线／菱形：3个seed等权均值；阴影／误差棒：±1个样本标准差；细线／圆点：各seed。
周期baseline曲线不平滑；训练长度作8个PPO轮的后向平滑，保留实际缺失点；均值仅在3个seed都有记录的轮次显示。
训练长度来自rank0、各自动态自对弈的已结束episode，不能当作固定对手能力；训练上限1024 tick，评估上限2000 tick。
F03只有best/final独立800局端点，不连线、不补造中途评估；周期baseline与独立端点使用不同评估协议。
S1原automatic=live、有效继承=recent8；final跨seed标准差仅相对改善1.79%，不代表强稳定性证据。

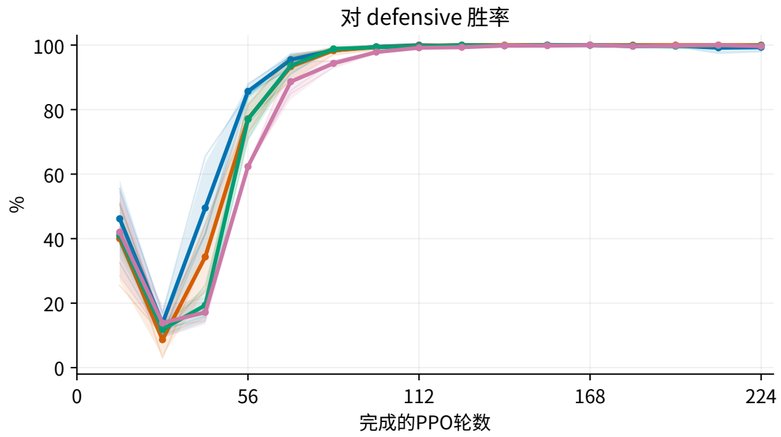
S2 自适应学习率：模型能力随训练变化

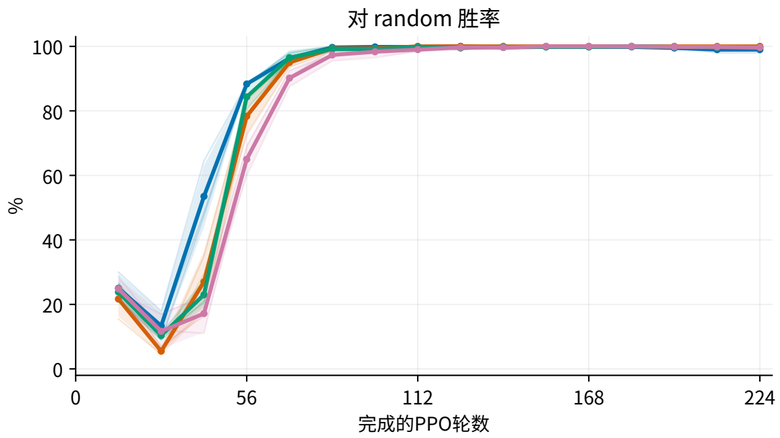
3个配对种子｜224轮PPO｜每轮256次逻辑Adam/rank｜周期评估每14轮一次、每次800局

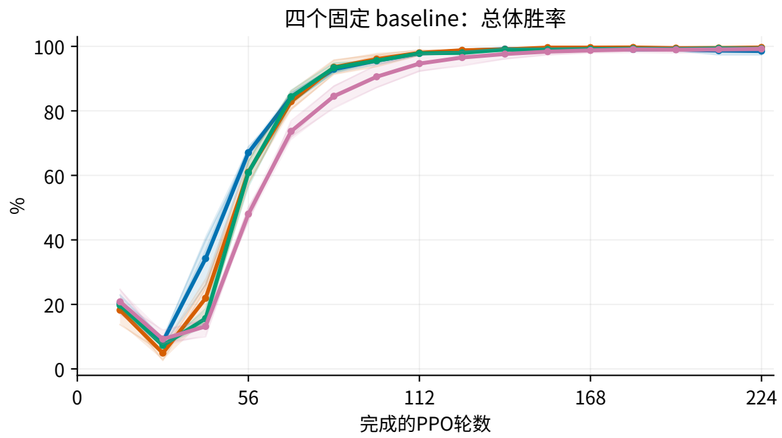
原范围 [1e-5,1e-2]
范围 [1e-5,1e-3] ★继承
critic固定1e-4
缩窄范围＋critic固定

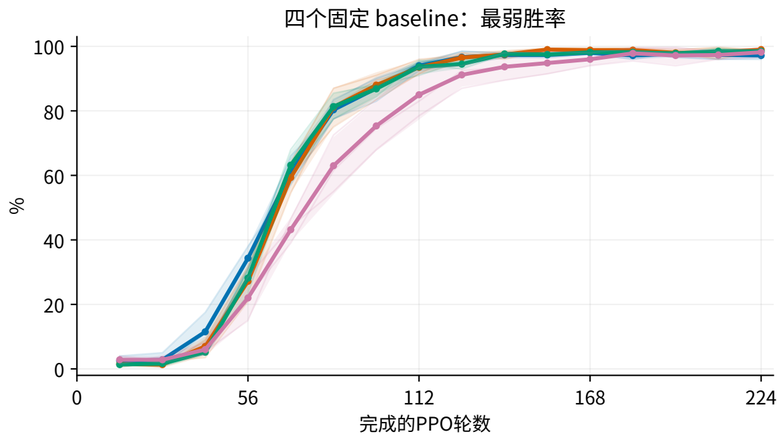


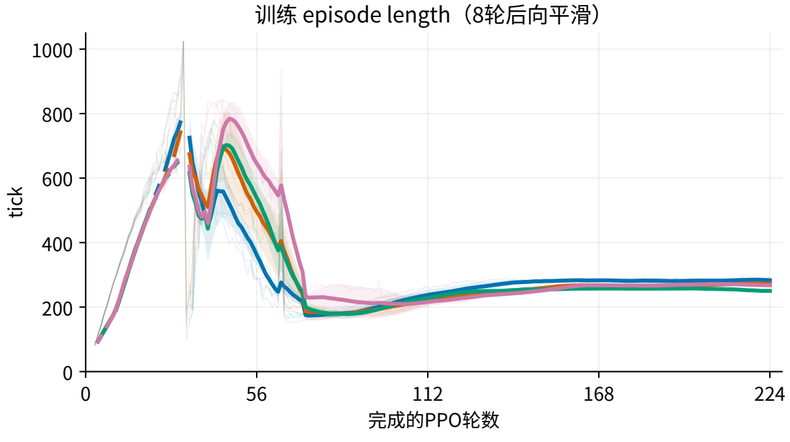


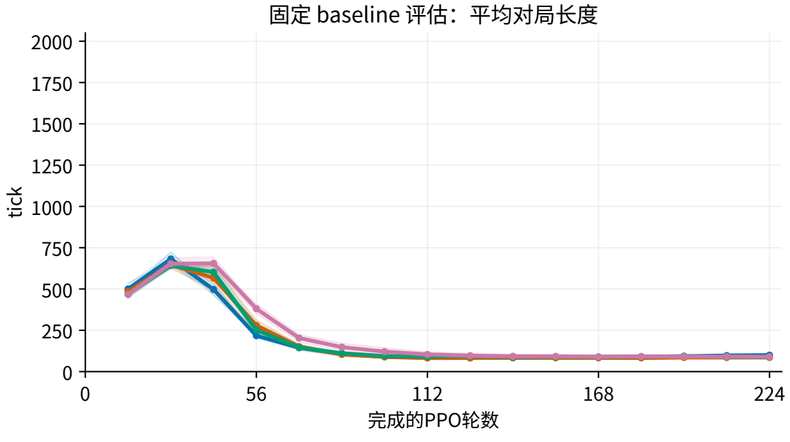


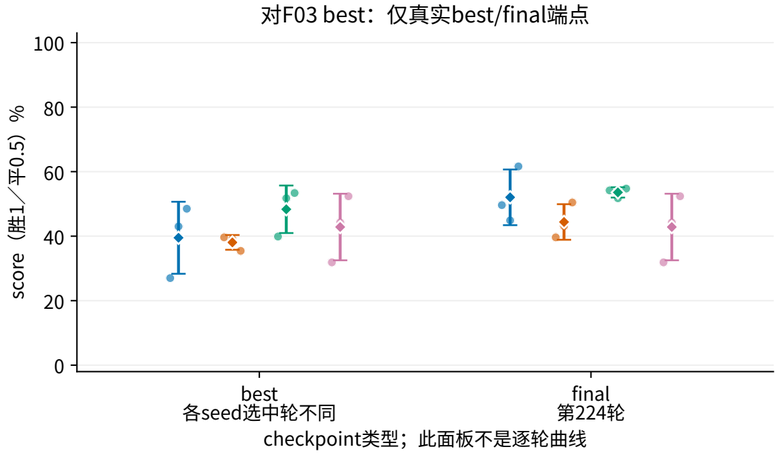










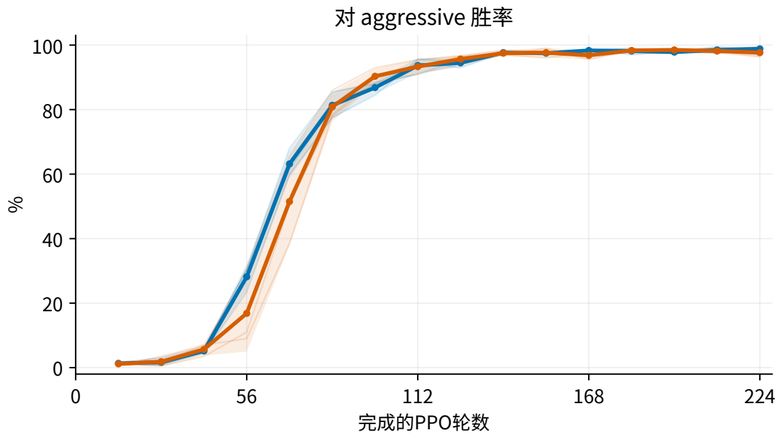


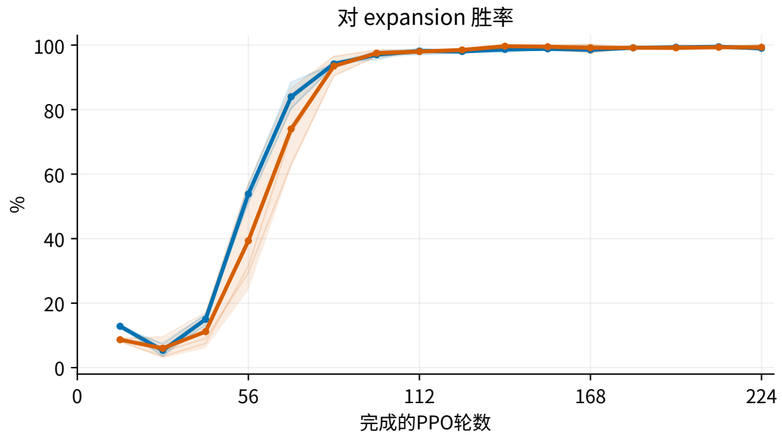
粗线／菱形：3个seed等权均值；阴影／误差棒：±1个样本标准差；细线／圆点：各seed。
周期baseline曲线不平滑；训练长度作8个PPO轮的后向平滑，保留实际缺失点；均值仅在3个seed都有记录的轮次显示。
训练长度来自rank0、各自动态自对弈的已结束episode，不能当作固定对手能力；训练上限1024 tick，评估上限2000 tick。
F03只有best/final独立800局端点，不连线、不补造中途评估；周期baseline与独立端点使用不同评估协议。

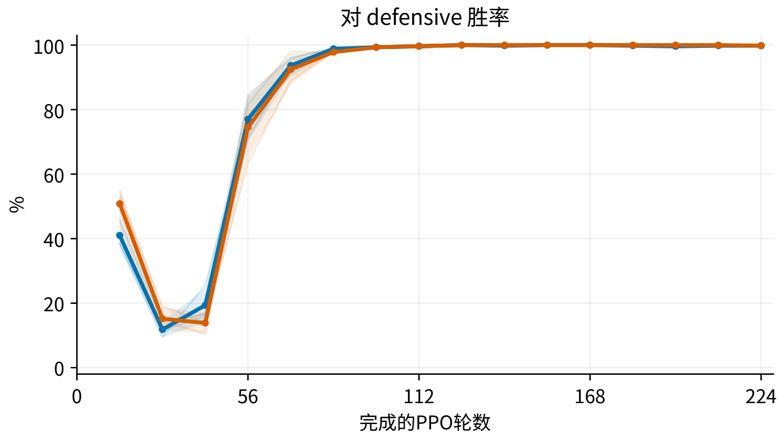
S3 目标KL：模型能力随训练变化

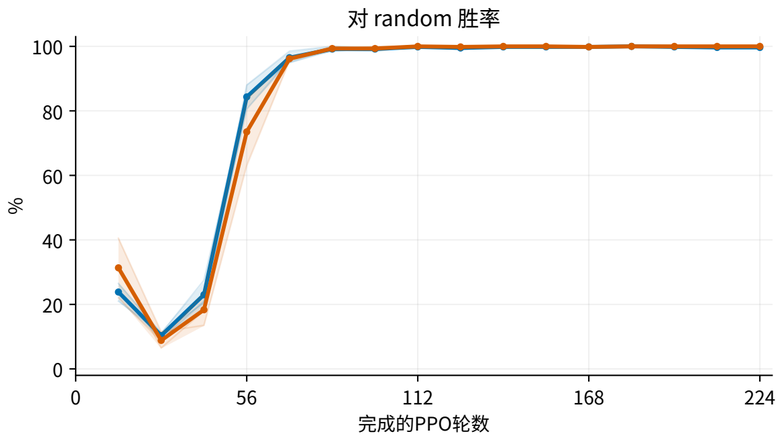
3个配对种子｜224轮PPO｜每轮256次逻辑Adam/rank｜周期评估每14轮一次、每次800局

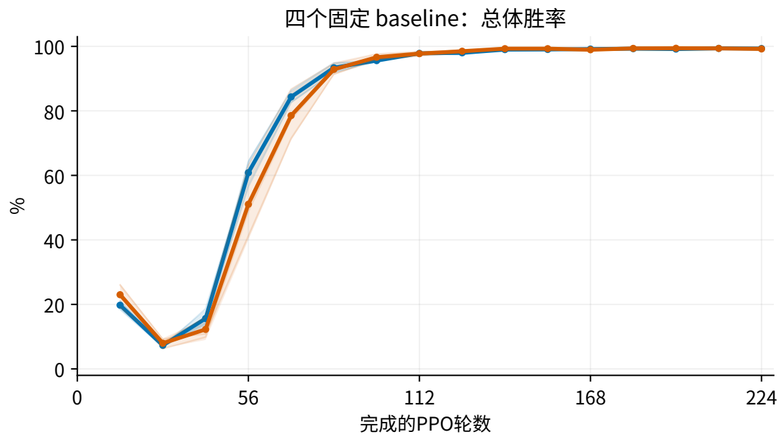
KL 0.01 ★继承　KL 0.02

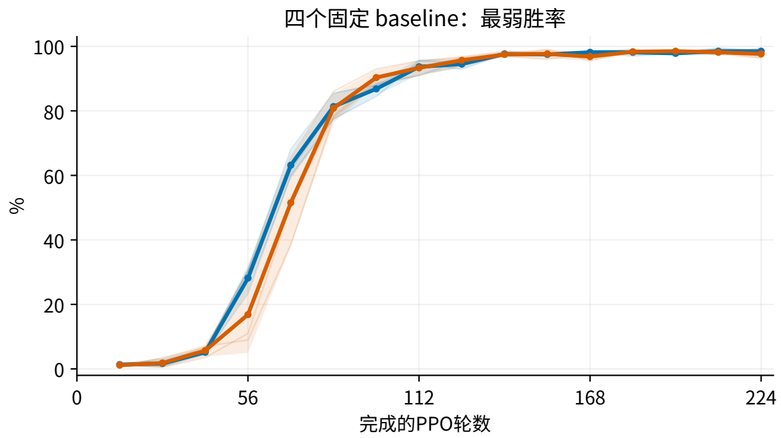


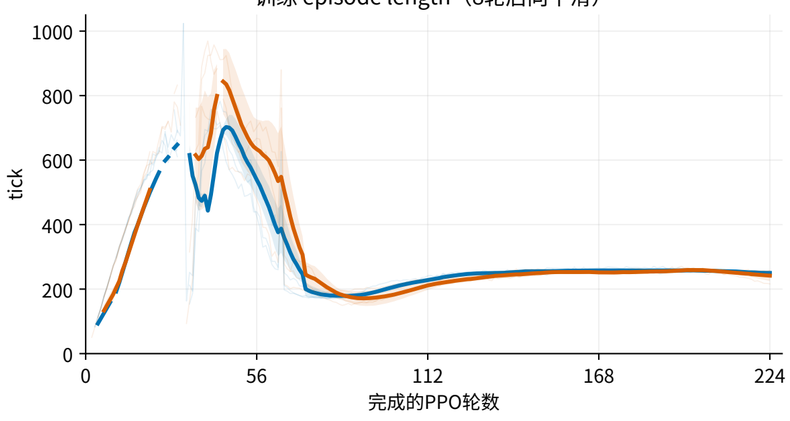


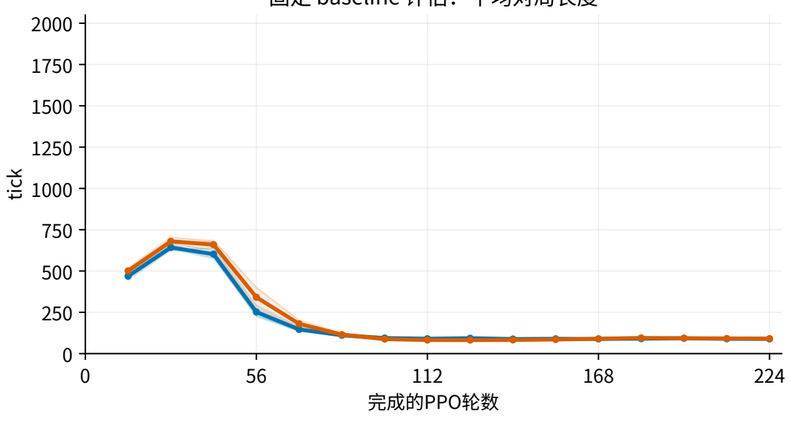


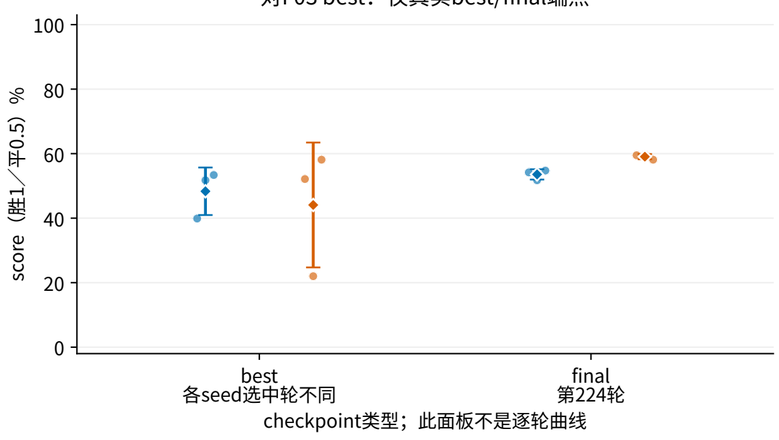










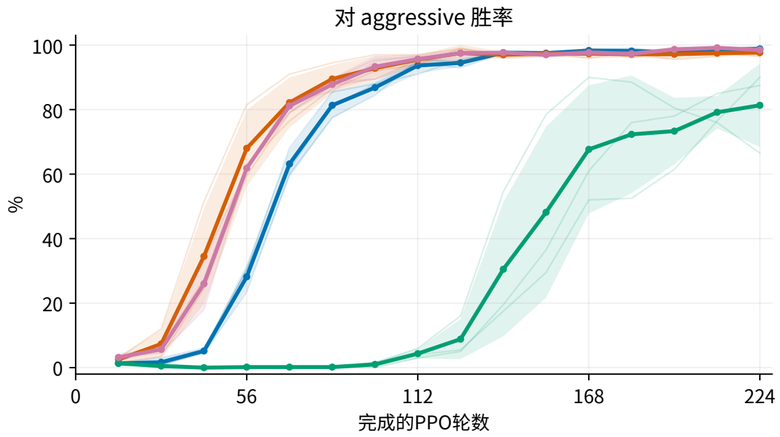


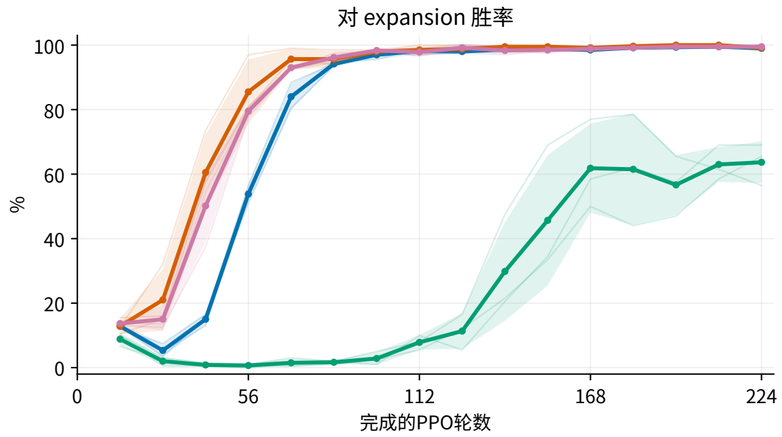
粗线／菱形：3个seed等权均值；阴影／误差棒：±1个样本标准差；细线／圆点：各seed。
周期baseline曲线不平滑；训练长度作8个PPO轮的后向平滑，保留实际缺失点；均值仅在3个seed都有记录的轮次显示。
训练长度来自rank0、各自动态自对弈的已结束episode，不能当作固定对手能力；训练上限1024 tick，评估上限2000 tick。
F03只有best/final独立800局端点，不连线、不补造中途评估；周期baseline与独立端点使用不同评估协议。

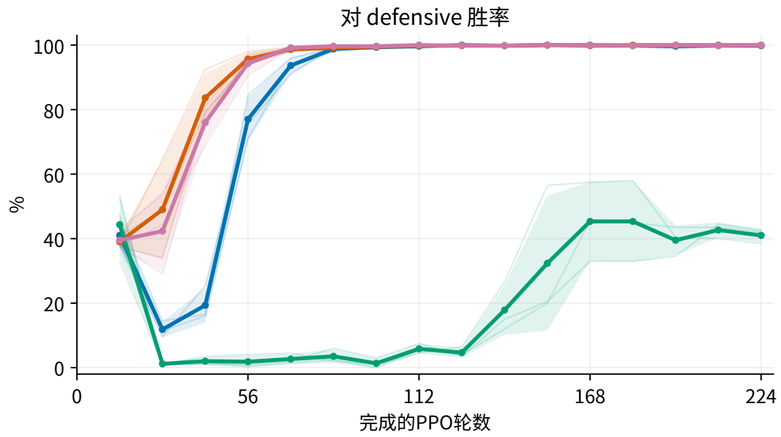
S4 折扣率与课程：模型能力随训练变化

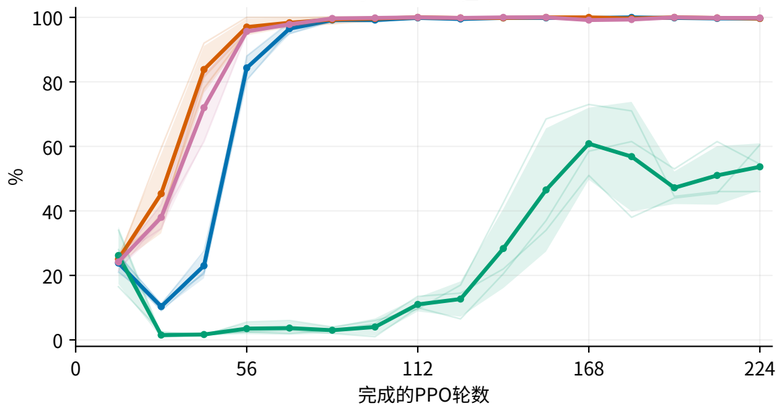
3个配对种子｜224轮PPO｜每轮256次逻辑Adam/rank｜周期评估每14轮一次、每次800局

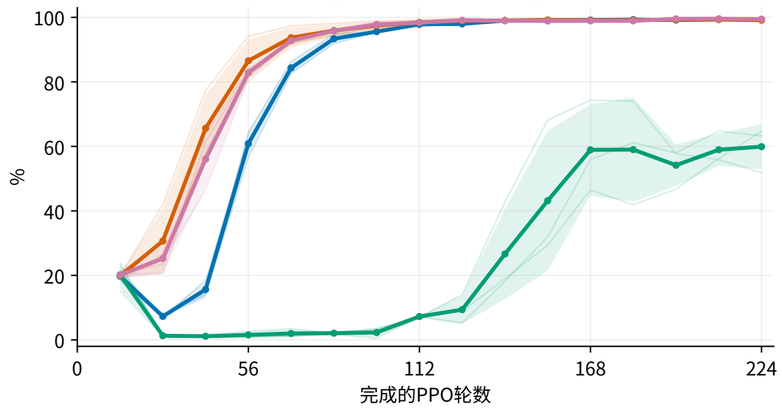
γ 0.995　γ 0.999
γ 0.99　γ 0.99→0.999 ★继承

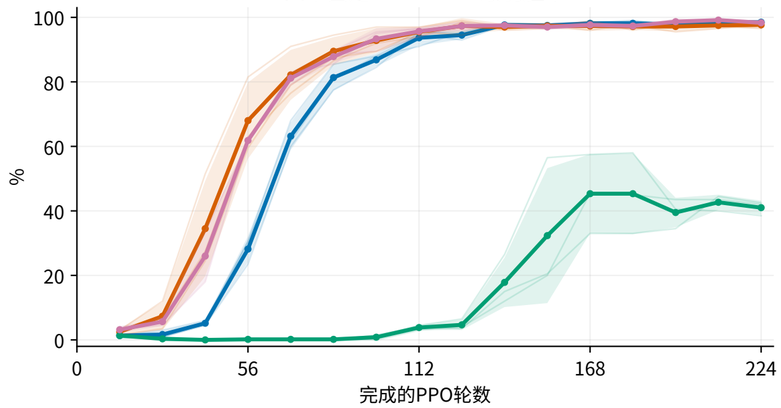


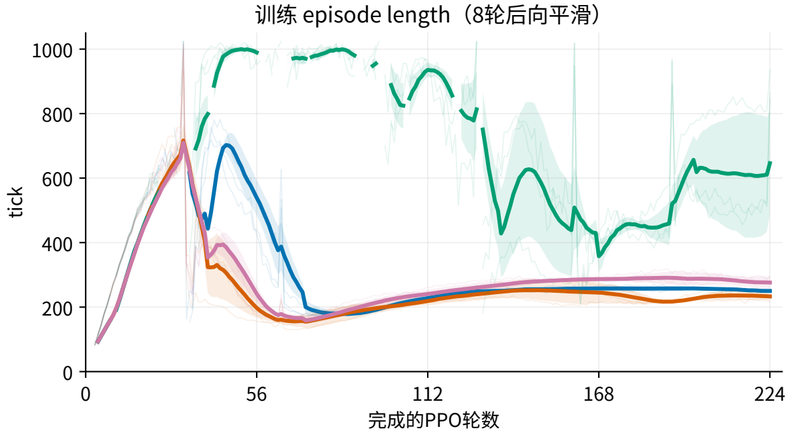


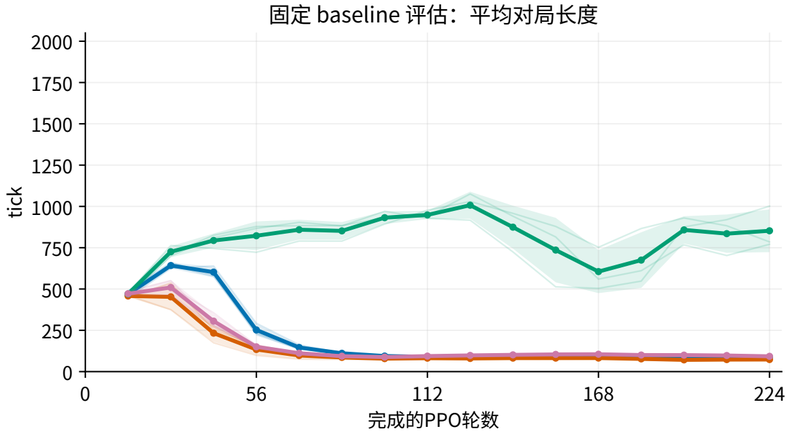


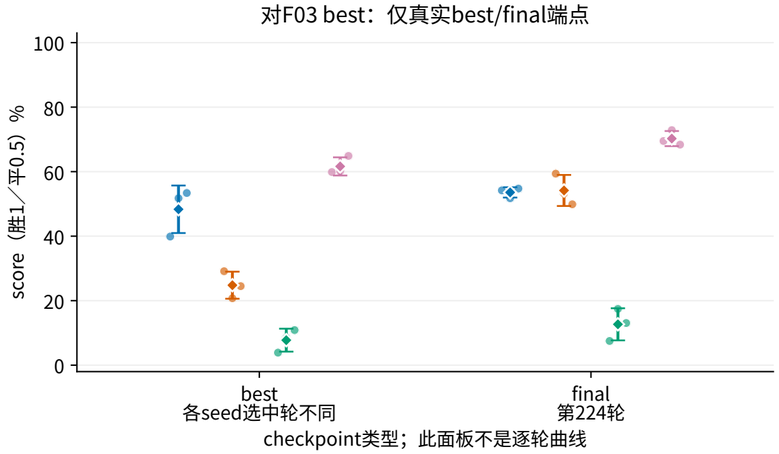










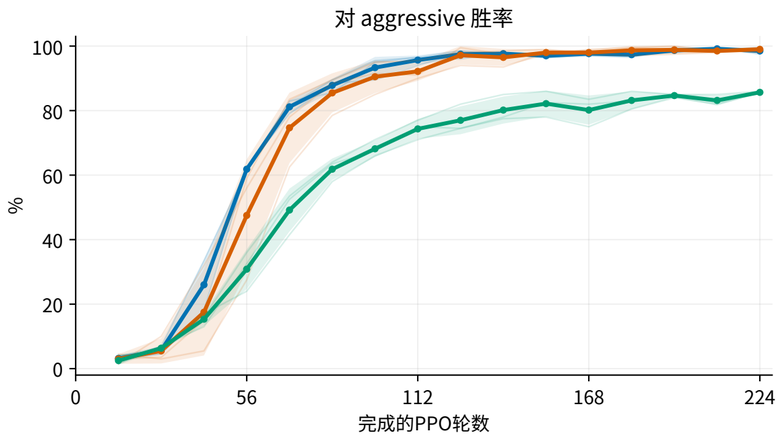


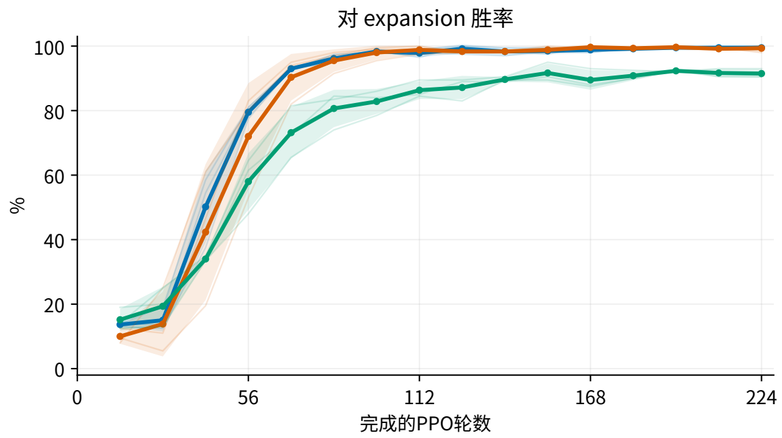
粗线／菱形：3个seed等权均值；阴影／误差棒：±1个样本标准差；细线／圆点：各seed。
周期baseline曲线不平滑；训练长度作8个PPO轮的后向平滑，保留实际缺失点；均值仅在3个seed都有记录的轮次显示。
训练长度来自rank0、各自动态自对弈的已结束episode，不能当作固定对手能力；训练上限1024 tick，评估上限2000 tick。
F03只有best/final独立800局端点，不连线、不补造中途评估；周期baseline与独立端点使用不同评估协议。

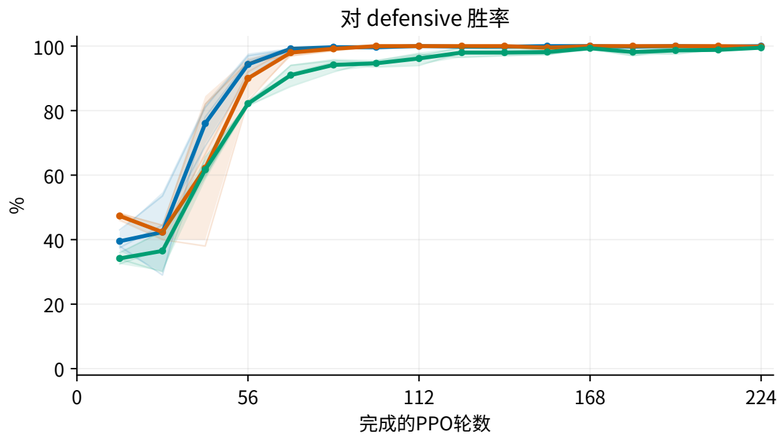
S5 ViT容量与空间模式：模型能力随训练变化

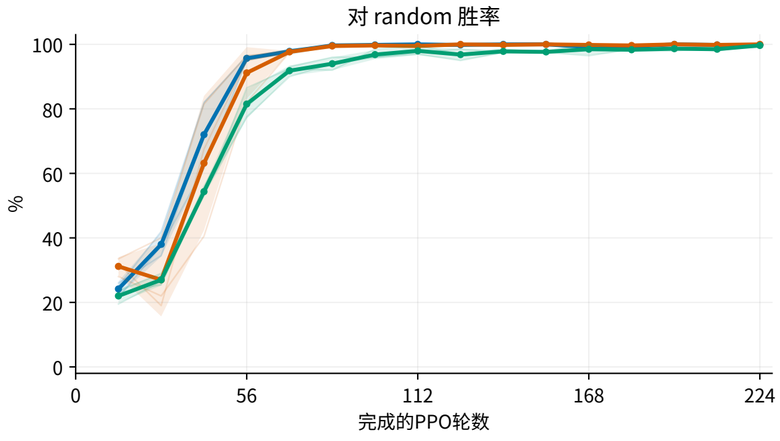
3个配对种子｜224轮PPO｜每轮256次逻辑Adam/rank｜周期评估每14轮一次、每次800局

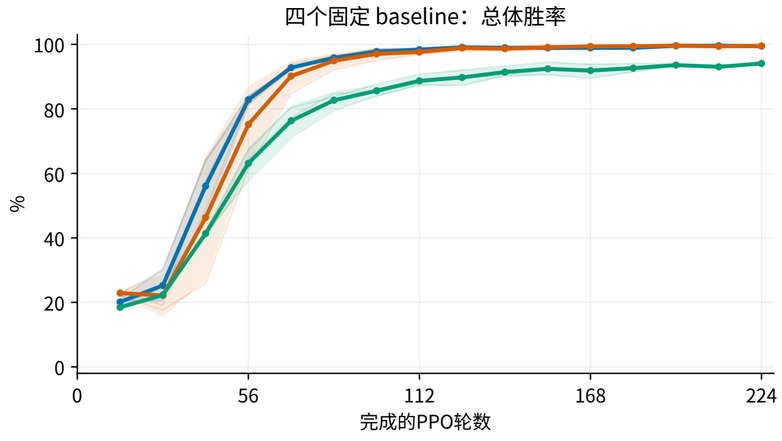
ViT1M dense ★继承
ViT10M dense
ViT10M pyramid

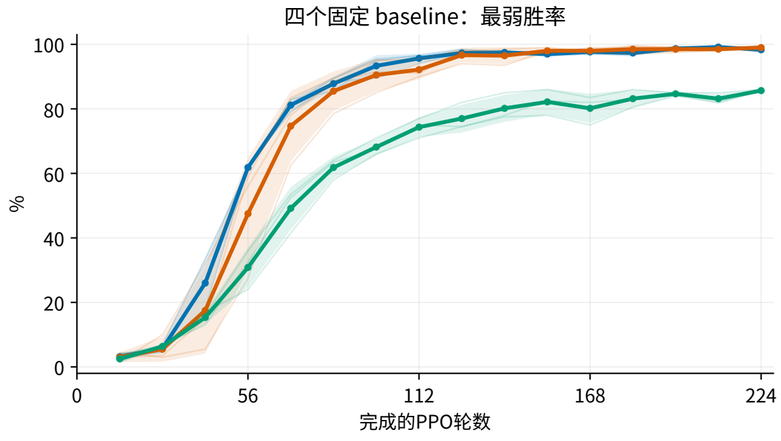


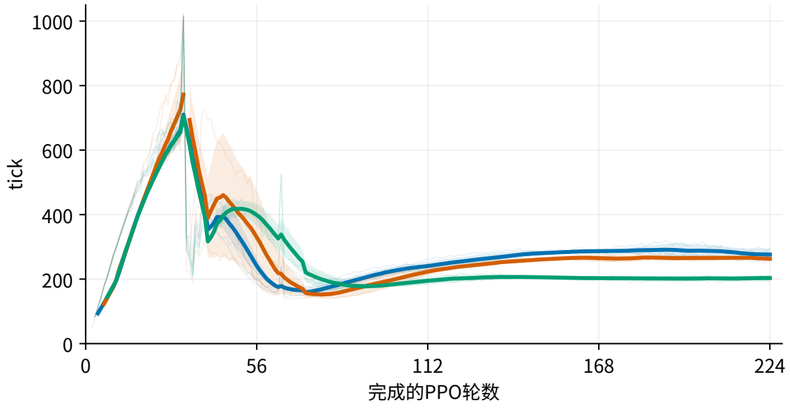


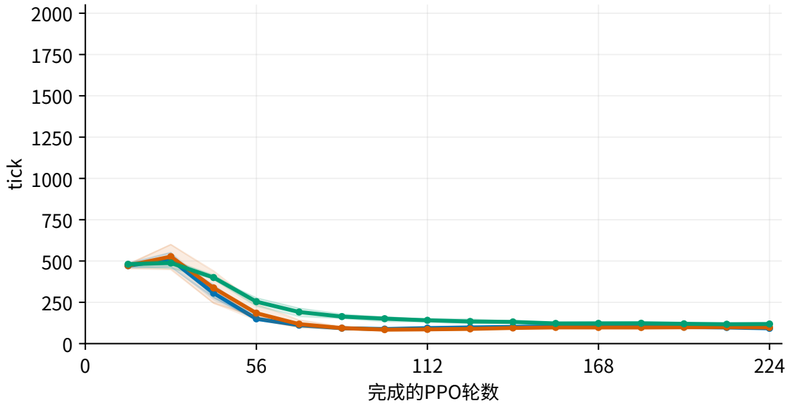


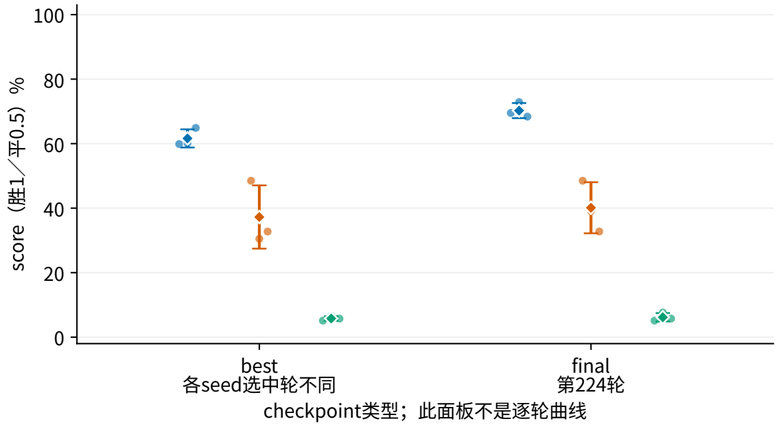










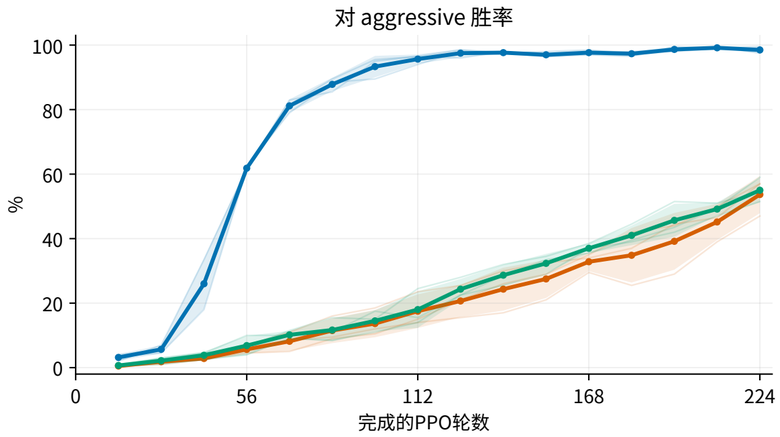


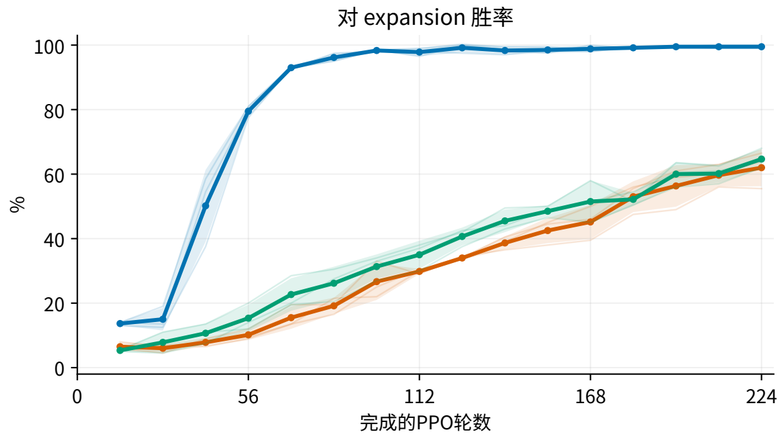
粗线／菱形：3个seed等权均值；阴影／误差棒：±1个样本标准差；细线／圆点：各seed。
周期baseline曲线不平滑；训练长度作8个PPO轮的后向平滑，保留实际缺失点；均值仅在3个seed都有记录的轮次显示。
训练长度来自rank0、各自动态自对弈的已结束episode，不能当作固定对手能力；训练上限1024 tick，评估上限2000 tick。
F03只有best/final独立800局端点，不连线、不补造中途评估；周期baseline与独立端点使用不同评估协议。

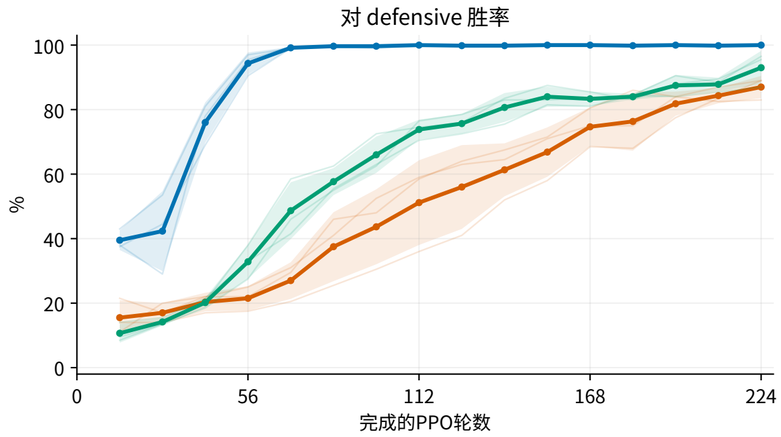
S6 ResNet与ViT：模型能力随训练变化

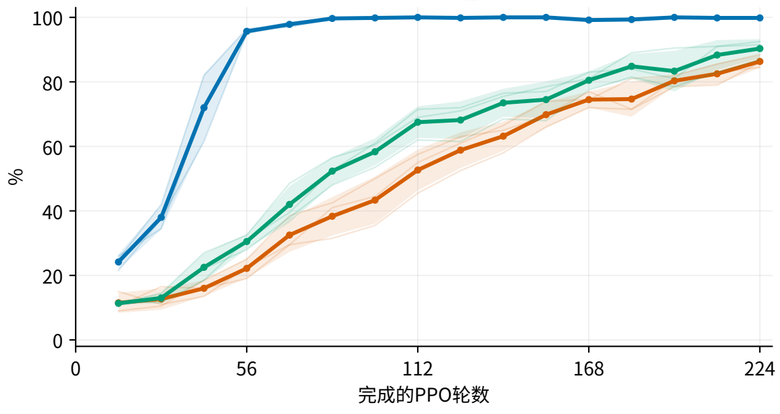
3个配对种子｜224轮PPO｜每轮256次逻辑Adam/rank｜周期评估每14轮一次、每次800局

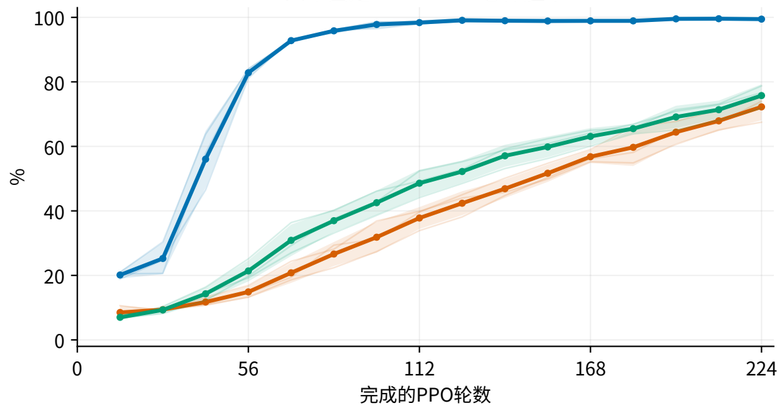
ViT1M dense ★继承
ResNet1M pyramid
ResNet1M dense

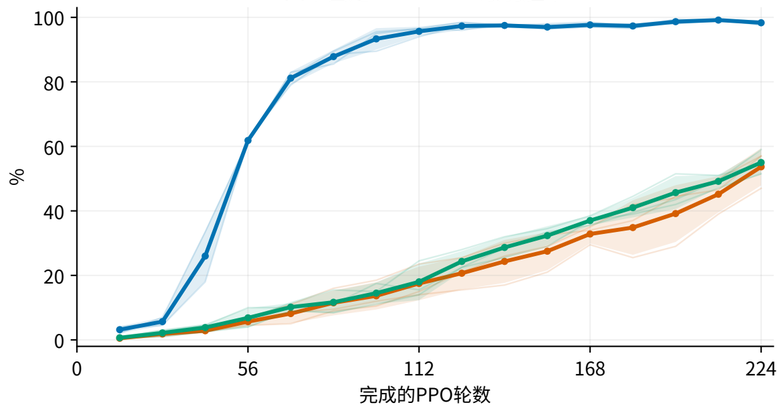


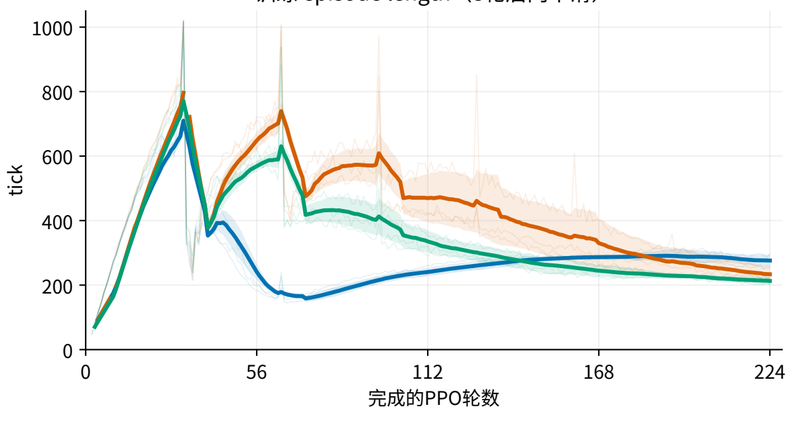


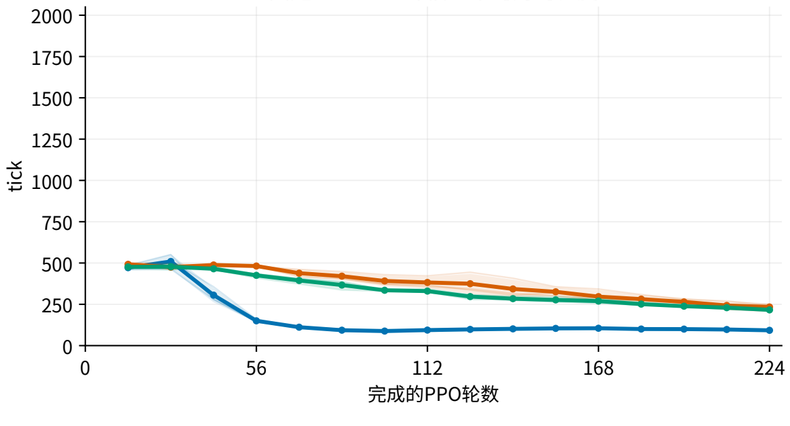


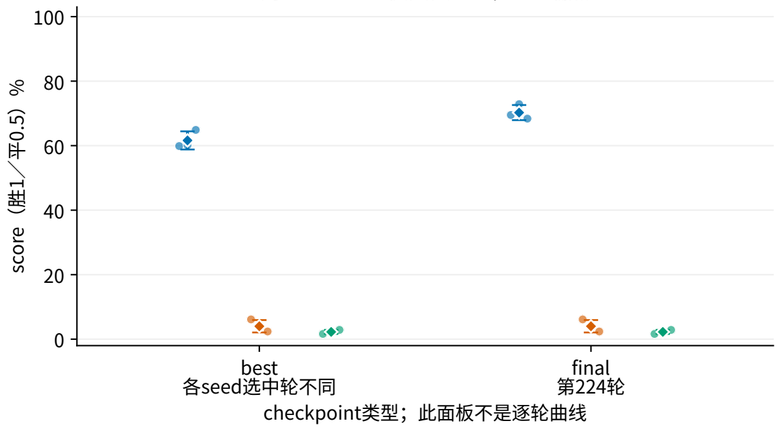








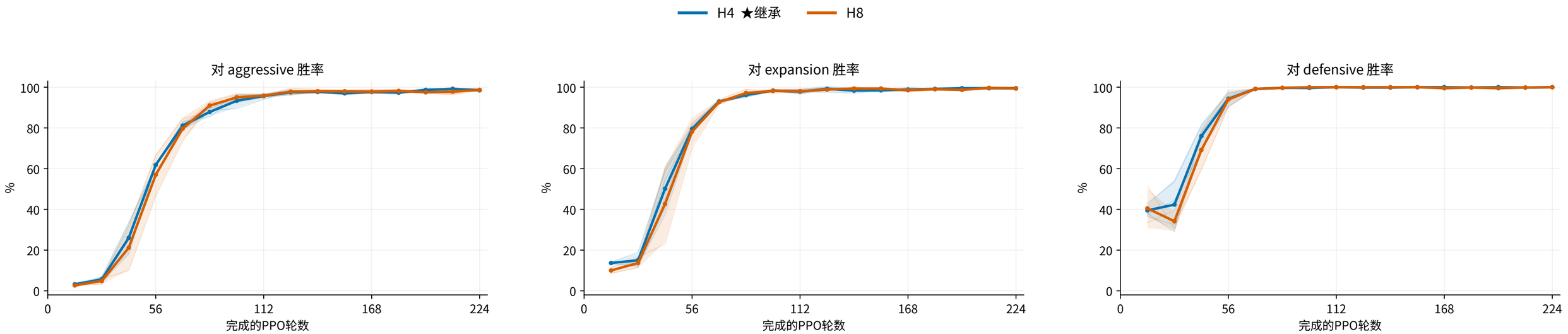




粗线／菱形：3个seed等权均值；阴影／误差棒：±1个样本标准差；细线／圆点：各seed。
周期baseline曲线不平滑；训练长度作8个PPO轮的后向平滑，保留实际缺失点；均值仅在3个seed都有记录的轮次显示。
训练长度来自rank0、各自动态自对弈的已结束episode，不能当作固定对手能力；训练上限1024 tick，评估上限2000 tick。
F03只有best/final独立800局端点，不连线、不补造中途评估；周期baseline与独立端点使用不同评估协议。

S7 观察历史长度：模型能力随训练变化

3个配对种子｜224轮PPO｜每轮256次逻辑Adam/rank｜周期评估每14轮一次、每次800局



粗线／菱形：3个seed等权均值；阴影／误差棒：±1个样本标准差；细线／圆点：各seed。
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